

John Hu

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Residential Address: Durham, NC, 27708

OBJECTIVE

Rising Junior at Duke University keen on obtaining an **internship** in the EE industry where I can leverage my strong **engineering, prototyping, programming, and communication skills** to contribute to team success

HIGHLIGHT OF QUALIFICATIONS

- Internship & project experience in e2e engineering, full stack development, and machine learning
- Hands-on experience in mechanical design & engineering with **Fusion360**, soldering and crimping, circuit design using **KiCad/Altium**, testing with oscilloscope, LA, signal generators.
- Solid knowledge of signal processing, motor control algorithms, data structures, simulation, and analog circuitry
- Proficient at **Python (PyTorch, NumPy)**, **C**, **C++**, **Java**, **Linux**, **MIPS Assembly**, **Verilog**, **MATLAB/Simulink**, **Golang**, **React**, **MongoDB**
- Quick learner, problem-solver and hard worker; positive, detail-oriented and responsible with excellent teamwork and communication skills; flexible with strong curiosity and work ethics

EDUCATION AND TRAINING

Bachelor of Science in Engineering, Duke University, Durham, USA 09/2024 – 05/2028 (expected)

- Major: **Electrical and Computer Engineering**; GPA: **3.63**
- Main courses: *Microelectronics; Robotics; Linear Algebra; Signals and System; Computer Architecture*
- **Key project**: Developed an autonomous robotic pick-and-place system using **ROS2**, OpenCV-based object recognition, and inverse kinematic to simulate Kinova Gen3 robotic arm operations
- **Other Certification**: Duke Machine Shop certified

PROFESSIONAL EXPERIENCE

Research Assistant, Duke University, Kotwal Lab, Durham, USA 08/2026 – Present

- Conducted experiments, processed and analyzed data using laboratory techniques and software to support research objectives in the field of Particle Accelerators

AI Engineer (Intern), Fast Approach Inc., Toronto, Canada 05/2026 – 07/2026

- Analyzed 20M+ advertising data using **Python (Pandas, Matplotlib, NumPy)** to identify campaign trends, customer behavior, and performance insights
- Built and evaluated multiple machine learning classification models to predict advertisement click-through rates, achieving an F1 score of 0.75 through feature engineering and model optimization
- Designed and developed a no-code GUI application to automate data cleaning and preprocessing, reducing processing time by 80%+ and improving workflow efficiency

Teaching Assistant, Duke University, Durham, USA 09/2025 – 12/2025

- Guided 15+ first-year engineering students in Python programming, debugging, and algorithmic thinking; Evaluated assignments, debugged student code, and monitored engineering programming projects' progress

PROJECT EXPERIENCE

Cycloidal Drive Proof of Concept Project 01/2026 – 03/2026

- Designed, modeled and optimized a compact cycloidal gearbox for robotic actuators by evaluating gear lobe count, eccentricity, bearing radius, and transmission efficiency through iterative prototyping and testing

Mechanical Keyboard Design Project 11/2025 – 12/2025

- Designed, assembled, and programmed a fully customized mechanical keyboard featuring USB 2.0 connectivity, rotary encoder, per-key RGB lighting and QMK firmware running on an AVR ATmega32U4 microcontroller

Team Canada Member, Bronze Medal, International Young Physicists Tournament, Romania 04/2022 – 07/2022

- Selected as 1 of 5 members to represent Canada at the 35th International Young Physicists Tournament; Engineered high-fidelity mechanics simulations in MATLAB and rotating cylinder fluid simulations in ANSYS Fluent

EXTRACURRICULUM ACTIVITIES

- President, *Duke Hyperloop*, Durham, USA 08/2026 – Present
- Electrical Lead, *Duke Hyperloop*, Durham, USA 08/2025 – 05/2026
- **Interests**: running, swimming, video editing, music
- **Languages**: Bilingual – English & Mandarin, basic in French, German